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Centre Number

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Student Number

**TRIAL** HIGHER SCHOOL CERTIFICATE EXAMINATION  
(BOSS version)



## Mathematics Extension 2

### General

### Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page

### Total marks:

100

### Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

### Section II – 90 marks (pages 7–17)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

#### Format Note:

Follows the 2026 exam format and 2017 NESA syllabus (not 2024).

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<https://vumaths.com/>

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Consider the line segments in 3D space given by the vector equations below.

$$\vec{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, 0 \leq \lambda \leq 3; \quad \vec{s} = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 2 \\ -4 \end{pmatrix}, -1 \leq \mu \leq 0.5; \quad \vec{v} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}, 0 \leq \gamma \leq 2$$

Which of the following statements about the line segments is NOT true?

- A. At least two of them are parallel
- B. At least two of them are perpendicular
- C. At least two of them intersect
- D. At least two of them have the same length

2 Consider the statement:

$$\forall n \in \mathbb{Z}, n^3 \text{ is even.}$$

Which of the following is the negation of the statement?

- A.  $\exists n \in \mathbb{Z}$ , such that  $n^3$  is odd.
- B.  $\exists n \in \mathbb{Z}$ , such that  $n^3$  is even.
- C.  $\forall n \in \mathbb{Z}, n^3$  is odd.
- D.  $n^3$  is odd  $\implies n \in \mathbb{Z}$

3 A complex number  $z$  is defined such that  $|z - 3 - 4i| = 2$ . Which of the following is the maximum value of  $|z - i|$ ?

- A.  $3\sqrt{2} - 2$
- B.  $3\sqrt{2} + 2$
- C. 5
- D. 7

- 4 A particle is moving in simple harmonic motion along the  $x$ -axis. Its velocity  $v \text{ ms}^{-1}$  at displacement  $x$  metres is given by the equation  $v^2 = 9 + 16x - 4x^2$ .

Which of the following correctly identifies the period of the motion and the maximum distance of the particle from the origin?

- A. Period  $\pi$ , Maximum distance  $\frac{5}{2}$  m
- B. Period  $2\pi$ , Maximum distance  $\frac{5}{2}$  m
- C. Period  $\pi$ , Maximum distance  $\frac{9}{2}$  m
- D. Period  $2\pi$ , Maximum distance  $\frac{9}{2}$  m
- 5 Consider the logical proposition  $P$ :  
“For all real numbers  $x$ , there exists a real number  $y$  such that if  $x > 0$ , then  $xy < 0$ .”
- Which of the following represents the correct logical negation of proposition  $P$ ?
- A. There exists a real number  $x$  such that for all real numbers  $y$ , if  $x > 0$ , then  $xy \geq 0$ .
- B. For all real numbers  $x$  and for all real numbers  $y$ ,  $x \leq 0$  or  $xy \geq 0$ .
- C. There exists a real number  $x$  such that for all real numbers  $y$ ,  $x > 0$  **and**  $xy \geq 0$ .
- D. There exists a real number  $x$  such that there exists a real number  $y$  where  $x > 0$  **and**  $xy \geq 0$ .
- 6 The complex numbers  $z_1$  and  $z_2$  satisfy the following conditions:

$$z_1 + z_2 = 6 + 8i$$

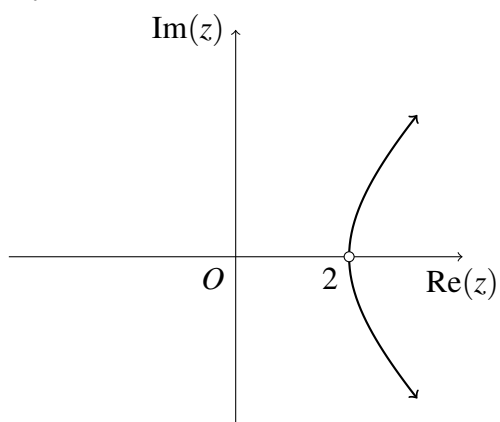
$$|z_1| + |z_2| = 14$$

What is the minimum value of  $|z_1 - z_2|$ ?

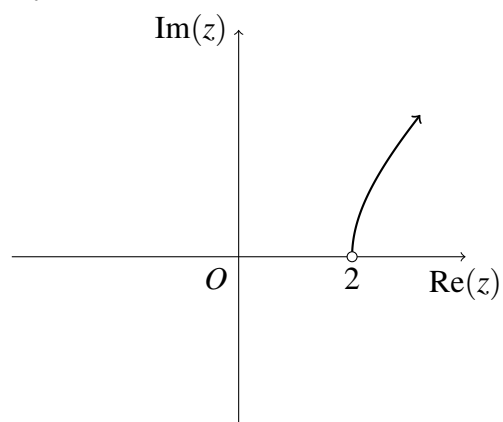
- A.  $2\sqrt{6}$
- B.  $4\sqrt{6}$
- C. 10
- D. 14

7 Which of the following Argand diagrams represents the locus of the complex number  $z$  that satisfies the equation  $\text{Arg}(z-2) + \text{Arg}(z+2) = \frac{\pi}{2}$ ?

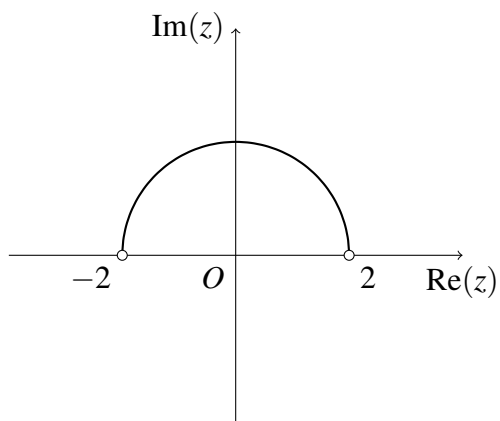
A.



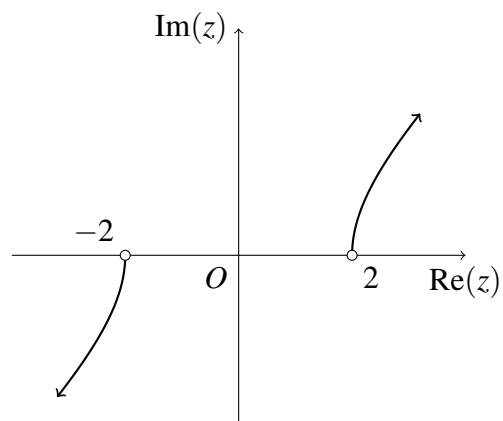
B.



C.

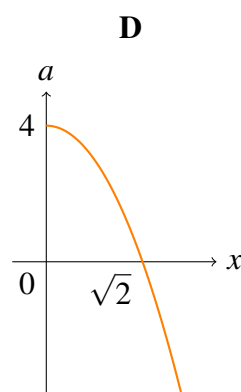
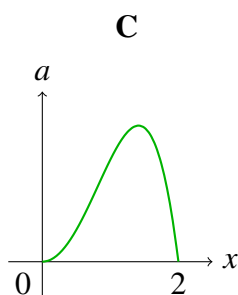
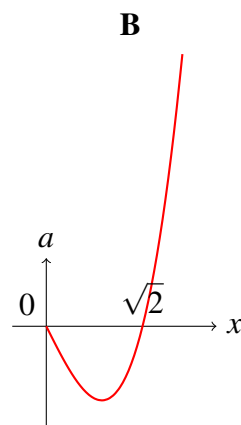
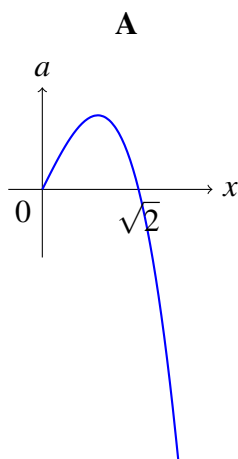


D.



- 8 A particle moves in a straight line such that its velocity  $v$  at displacement  $x$  is given by  $v^2 = 4x^2 - x^4$ , for  $0 \leq x \leq 2$ .

We want to represent the acceleration  $a$  of the particle as a function of its displacement  $x$ . Which colour do you like?



- 9 Let  $S$  be a sphere in 3-dimensional space defined by the vector equation  $|\mathbf{r} - \mathbf{c}| = R$ , where  $\mathbf{c}$  is the position vector of its centre and  $R$  is its radius.

A line  $L$  is defined by the vector equation  $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$ , where  $\mathbf{d}$  is a non-zero direction vector.

It is given that:

- A.  $(\mathbf{a} - \mathbf{c}) \cdot \mathbf{d} = 0$
- B.  $0 < |\mathbf{a} - \mathbf{c}| < R$

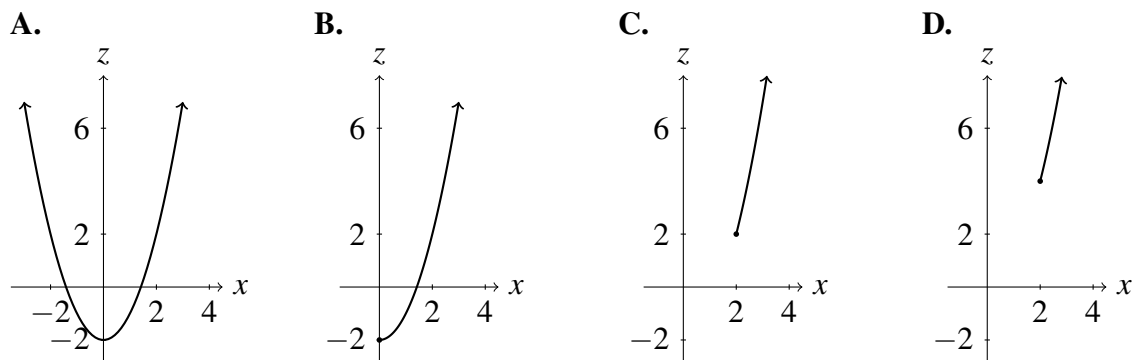
Which of the following best describes the geometric scenario between the sphere  $S$ , the line  $L$ , and the point  $\mathbf{a}$ ?

- A. The point  $\mathbf{a}$  is the midpoint of a chord formed by  $L$  intersecting  $S$ .
- B. The line  $L$  is tangent to the sphere  $S$  at the point  $\mathbf{a}$ .
- C. The point  $\mathbf{a}$  is an endpoint of a chord formed by  $L$  intersecting  $S$ .
- D. The line  $L$  passes directly through the centre  $\mathbf{c}$  of the sphere.

- 10 The curve  $C$  is defined by the following parametric equations for all real values of  $t$ :

$$x(t) = e^t + e^{-t}, \quad y(t) = \sin(t), \quad z(t) = e^{2t} + e^{-2t}$$

One of the following graphs represents the projection of the curve  $C$  onto the  $xz$ -plane. What character do you like the most. "A" for Awesome, "B" for Beautiful, "C" for Cute, "D" for Dazzling.



## Section II

**90 marks**

**Attempt Questions 11–16**

**Allow about 2 hours and 45 minutes for this section**

Answer the questions in the Section II Writing Booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

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### Question 11 (14 marks)

- (a) By considering the roots of  $z^5 = 16\sqrt{2}(1 - i)$ , show that: **2**

$$\cos \frac{\pi}{20} + \cos \frac{7\pi}{20} + \cos \frac{9\pi}{20} + \cos \frac{17\pi}{20} = \frac{\sqrt{2}}{2}$$

- (b) Let  $z$  be a non-zero complex number such that  $\left|z - \frac{2i}{z}\right| = 1$ . **2**  
By using the triangle inequality, or otherwise, find the greatest possible value of  $|z|$ .

- (c) Let  $z = 1 - i\sqrt{3}$  and  $w = -1 + i$ . **2**

Find the smallest positive integer  $n$  such that  $(zw)^n$  is a purely imaginary number.

**Question 11 continues on page 8**

## Question 11 (continued)

- (d) Let  $\underline{p}$  and  $\underline{q}$  be two non-zero 3-dimensional vectors. 3

By considering the fact that  $|\lambda \underline{p} + \underline{q}|^2 \geq 0$  for all real scalar values of  $\lambda$ , prove that:

$$(\underline{p} \cdot \underline{q})^2 \leq |\underline{p}|^2 |\underline{q}|^2$$

- (e) Let  $z$  be a complex number such that  $|z| = 1$  and  $z \neq -1$ . 2

Prove that  $\frac{z-1}{z+1}$  is purely imaginary.

- (f) Find real constants  $A$ ,  $B$ , and  $C$  such that, for all  $u$ , 3

$$\frac{1}{u^3 + u^2 + u + 1} = \frac{A}{u+1} + \frac{Bu+C}{u^2+1}$$

Hence, evaluate

$$\int \frac{\sec^2 x}{\tan^3 x + \tan^2 x + \tan x + 1} dx$$

**End of Question 11**



**Question 12** (16 marks)

- (a) Evaluate the following definite integral:

**3**

$$\int_0^{\frac{\pi}{4}} \frac{\sec^4 x}{(1 + \tan x)^3} dx$$

- (b) A sequence is defined recursively by
- $v_1 = \sqrt{3}$
- and for all integers
- $n \geq 1$
- :

**3**

$$v_{n+1} = \frac{v_n}{1 + \sqrt{1 + v_n^2}}$$

Prove by mathematical induction that for all integers  $n \geq 1$ ,

$$v_n = \tan\left(\frac{\pi}{3 \cdot 2^{n-1}}\right)$$

- (c) On an Argand diagram, sketch the region representing the set of all
- $z$
- satisfying

**3**

$$1 \leq |z - 2i| \leq 3 \text{ and } \frac{\pi}{6} \leq \arg(z - 2i) \leq \frac{5\pi}{6},$$

showing the coordinates of any vertices in Cartesian form.

Hence, or otherwise, determine the maximum value of  $|z|$  for any complex number  $z$  in this region.

- (d) (i) Express  $P(x) = \frac{1}{x(1+x^2)^2}$  in the fractional form  $\frac{A}{x} + \frac{Bx+C}{1+x^2} + \frac{Dx+E}{(1+x^2)^2}$ ,  
where  $A, B, C, D, E$  are constants. **2**

- (ii) Hence, find  $\int \frac{1}{x(1+x^2)^2} dx$ . **2**

- (e) Suppose a particle is moving horizontally along the  $x$ -axis. Initially, the particle is at a displacement of  $x = 2$  metres from the origin and moving with a velocity of  $3 \text{ ms}^{-1}$ . The acceleration of the particle is given by  $\ddot{x} = 4x - \frac{4}{x^3}$  where  $x > 0$ .

- (i) Show that the velocity of the particle is given by  $\dot{x} = 2 \left( x - \frac{1}{x} \right)$ . **1**

- (ii) Hence, find an expression for  $x^2$  as a function of  $t$ . **2**

**End of Question 12**

**Question 13** (15 marks)

- (a) Let
- $a > 0$
- . Evaluate the integral:

**3**

$$\int_0^a \frac{\ln(1+ax)}{1+x^2} dx$$

(You may use the property that  $\int_0^c f(u) du = \int_0^c f(c-u) du$  without proof).

- (b) The points  $A$ ,  $B$ , and  $C$  have position vectors  $\vec{a}$ ,  $\vec{b}$ , and  $\vec{c}$  respectively, relative to an origin  $O$ . These four points form the vertices of a tetrahedron  $OABC$ .

- (i) The point  $P$  is the centroid (the point where the medians of the triangle intersect) of the face  $ABC$ . Let  $M$  be the midpoint of  $AB$ . Given that the centroid  $P$  divides the median  $CM$  internally in the ratio  $2 : 1$ , show that the position vector of  $P$  is given by **2**

$$\vec{p} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

- (ii) The point  $Q$  is the centroid of the face  $OBC$ . Prove that the line segment  $PQ$  is parallel to the edge  $OA$ , and find the ratio of the length of  $PQ$  to the length of  $OA$ . **2**

- (c) (i) Let  $f(x)$  and  $g(x)$  be continuous real-valued functions on the interval  $[a, b]$ . **2**  
By considering  $\int_a^b (tf(x) + g(x))^2 dx \geq 0$  for all real values of  $t$ , prove the Cauchy-Schwarz inequality for definite integrals:

$$\left( \int_a^b f(x)g(x) dx \right)^2 \leq \left( \int_a^b (f(x))^2 dx \right) \left( \int_a^b (g(x))^2 dx \right)$$

- (ii) Let  $h(x)$  be a continuously differentiable real-valued function on the interval  $[0, 1]$  such that  $h(1) = 0$ . By considering the derivative of  $x(h(x))^2$ , show that: **1**

$$\int_0^1 (h(x))^2 dx = -2 \int_0^1 xh(x)h'(x) dx$$

- (iii) Hence, use the inequality from part (i) to prove that: **1**

$$\int_0^1 (h'(x))^2 dx \geq \frac{1}{4} \int_0^1 (h(x))^2 dx$$

**Question 13 continues on page 11**

## Question 13 (continued)

(d) For integers  $m > 0$  and  $n > 0$ , we define the function  $B(m, n)$  by the definite integral:

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

(i) Use the substitution  $x = \frac{1}{1+u}$  to show that **2**

$$B(m, n) = \int_0^\infty \frac{u^{n-1}}{(1+u)^{m+n}} du.$$

Note: An integral over an infinite interval is evaluated using a limit,  
such that  $\int_0^\infty f(u) du = \lim_{R \rightarrow \infty} \int_0^R f(u) du$ .

(ii) Hence, or otherwise, evaluate **2**

$$\int_0^\infty \frac{u^2}{(1+u)^6} du.$$

**End of Question 13**

**Question 14** (15 marks)

(a) Let  $I_n = \int_0^1 \frac{1}{(x^2 + 1)^n} dx$  for integers  $n \geq 1$ .

(i) Show that  $2nI_{n+1} = 2^{-n} + (2n - 1)I_n$ . 2

(ii) Hence, evaluate  $\int_0^1 \frac{x^2}{(x^2 + 1)^3} dx$ . 2

(b) A particle moves along a straight line such that its acceleration  $a \text{ ms}^{-2}$  is given by

$$a = -3x^2v^3$$

where  $v \text{ ms}^{-1}$  is the velocity of the particle and  $x$  metres is its displacement from the origin  $O$ .

Initially, the particle is at a displacement of  $x = 1$  with a velocity of  $v = \frac{1}{2}$ .

(i) By expressing acceleration as  $v \frac{dv}{dx}$ , 2

show that the velocity of the particle is given by  $v = \frac{1}{x^3 + 1}$ .

(ii) Hence, find the exact time taken for the particle to reach the position  $x = 2$ . 2

(c) Let  $\triangle ABC$  be a triangle in the complex plane with vertices  $A, B, C$  represented by the distinct complex numbers  $a, b$ , and  $c$  respectively.

(i) Prove that  $\triangle ABC$  is equilateral if and only if  $a^2 + b^2 + c^2 = ab + bc + ca$ . 2

(ii) The complex numbers  $a, b$ , and  $c$  are the roots of the cubic equation: 2

$$z^3 - 3pz^2 + 3qz - r = 0$$

where  $p, q$ , and  $r$  are complex constants.

Using the result from part (i), or otherwise, deduce that  $\triangle ABC$  is equilateral if and only if  $p^2 = q$ .

**Question 14 continues on page 13**

## Question 14 (continued)

- (d) A sequence of real numbers is defined by  $x_1 = 1$  and the recurrence relation: **3**

$$x_{n+1} = x_n + \frac{1}{x_n^2} \quad \text{for integers } n \geq 1$$

Prove by mathematical induction, or otherwise, that  $x_n > \sqrt[3]{3n}$  for all integers  $n \geq 2$ .

**End of Question 14**

**Question 15** (15 marks)

- (a) Let  $OABC$  be a tetrahedron in three-dimensional space. The vertices  $A, B$ , and  $C$  have position vectors  $\mathbf{a}, \mathbf{b}$ , and  $\mathbf{c}$  respectively, relative to the origin  $O$ .

You are given that the pair of opposite edges  $OA$  and  $BC$  are perpendicular, and the pair of opposite edges  $OB$  and  $AC$  are also perpendicular.

- (i) Show that the remaining pair of opposite edges,  $OC$  and  $AB$ , are perpendicular. 2

- (ii) Prove that the sum of the squares of the lengths of any pair of opposite edges is equal. 2  
That is, show that:

$$|\vec{OA}|^2 + |\vec{BC}|^2 = |\vec{OB}|^2 + |\vec{AC}|^2 = |\vec{OC}|^2 + |\vec{AB}|^2$$

- (b) A particle of unit mass moves along the  $x$ -axis. The particle experiences a restoring force directed towards the origin of magnitude  $4x$  Newtons, where  $x$  is the displacement from the origin in metres. It also experiences a resistive force of magnitude  $4v$  Newtons, where  $v$  is the velocity in m/s.

Initially, the particle is at the origin with a velocity of 5 m/s in the positive  $x$ -direction.

- (i) Show that the equation of motion can be written as the differential equation 1  
 $\ddot{x} + 4\dot{x} + 4x = 0$ .
- (ii) Verify that  $x = 5te^{-2t}$  is a solution to the differential equation 2  
and satisfies the initial conditions.
- (iii) Find the exact maximum displacement of the particle from the origin. 1

**Question 15 continues on page 15**

## Question 15 (continued)

- (c) (i) By considering  $S - zS$ , or otherwise, where **2**

$S = z + 2z^2 + 3z^3 + \cdots + nz^n$ , show that for  $z \neq 1$ :

$$\sum_{k=1}^n kz^k = \frac{nz^{n+2} - (n+1)z^{n+1} + z}{(z-1)^2}$$

- (ii) Let  $z = \cos \theta + i \sin \theta$ , where  $0 < \theta < 2\pi$ . Show that: **2**

$$\frac{z}{(z-1)^2} = -\frac{1}{4 \sin^2 \left( \frac{\theta}{2} \right)}$$

- (iii) Hence, deduce that: **3**

$$\sum_{k=1}^n k \cos(k\theta) = \frac{(n+1) \cos(n\theta) - n \cos((n+1)\theta) - 1}{4 \sin^2 \left( \frac{\theta}{2} \right)}$$

**End of Question 15**

**Question 16** (15 marks)

(a) Let  $z = \cos \alpha + i \sin \alpha$ , where  $\alpha$  is a real number and  $\sin \alpha \neq 0$ .

(i) Prove that  $\frac{1}{1 - z \cos \alpha} = 1 + i \cot \alpha$ . **2**

(ii) You are given that: **2**

$$\sum_{k=1}^{\infty} k w^k = \frac{w}{(1-w)^2}, \quad \text{for } |w| < 1, \text{ Do NOT prove this.}$$

By letting  $w = z \cos \alpha$ , deduce that:

$$\sum_{k=1}^{\infty} k \cos(k\alpha) \cos^k \alpha = -\cot^2 \alpha$$

and

$$\sum_{k=1}^{\infty} k \sin(k\alpha) \cos^k \alpha = \cot \alpha$$

(b) A mechanical carriage of mass  $m$  on a carnival ride is projected vertically upwards from ground level with an initial velocity  $U$ . While in motion, the carriage experiences a resistive force of magnitude  $mkv^2$ , where  $v$  is its velocity and  $k$  is a positive constant. The acceleration due to gravity is  $g$ .

(i) Show that the maximum height  $H$  reached by the carriage is given by: **2**

$$H = \frac{1}{2k} \ln \left( 1 + \frac{kU^2}{g} \right)$$

(ii) After reaching its maximum height, the carriage falls back to the ground. **3**

Let  $W$  be its velocity immediately before it strikes the ground. Show that:

$$\frac{1}{W^2} = \frac{1}{U^2} + \frac{k}{g}$$

**Question 16 continues on page 17**



## Question 16 (continued)

(c) Consider a triangle  $OAB$ , where the position vectors of  $A$  and  $B$  relative to the origin  $O$  are  $\vec{OA} = \mathbf{a}$  and  $\vec{OB} = \mathbf{b}$ . Let the lengths of the sides of the triangle be denoted by  $OA = a$ ,  $OB = b$ , and  $AB = c$ .

(i) Show that the vector  $\mathbf{v} = b\mathbf{a} + a\mathbf{b}$  is parallel to the internal angle bisector of  $\angle AOB$ . 2

(ii) Let  $P$  be the point where the internal angle bisector of  $\angle AOB$  intersects the side  $AB$ . 2

By expressing the position vector  $\vec{OP}$  as a scalar multiple of  $\mathbf{v}$ ,  
prove that  $P$  divides the line segment  $AB$  in the ratio  $a : b$ .

(iii) Let  $I$  be the incenter of  $\triangle OAB$  2

(the point of intersection of the triangle's internal angle bisectors).

By considering the internal angle bisector of  $\angle OAP$  in  $\triangle OAP$ , or otherwise,  
deduce that the position vector of  $I$  is given by:

$$\vec{OI} = \frac{b\mathbf{a} + a\mathbf{b}}{a + b + c}$$

**End of paper**

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# REFERENCE SHEET

## Mathematics Advanced

### Mathematics Extension 1

### Mathematics Extension 2

#### Measurement

##### Length

$$l = \frac{\theta}{360} \times 2\pi r$$


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##### Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a+b)$$


---

##### Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$


---

##### Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$


---

##### Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{For } ax^3 + bx^2 + cx + d = 0:$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$


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##### Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

#### Financial Mathematics

$$A = P(1+r)^n$$


---

##### Sequences and series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, r \neq 1$$

$$S = \frac{a}{1-r}, |r| < 1$$


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##### Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

**Trigonometric Functions**

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \cos A = \frac{\text{adj}}{\text{hyp}}, \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

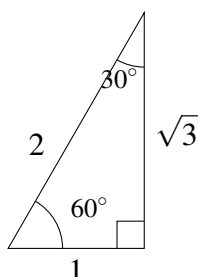
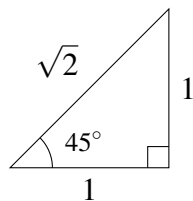
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$

**Trigonometric identities**

$$\sec A = \frac{1}{\cos A}, \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

**Compound angles**

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$$

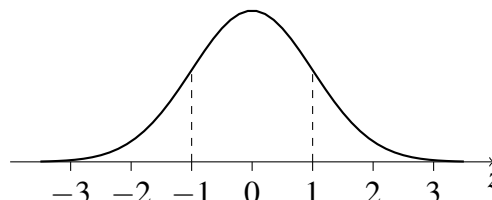
$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

**Statistical Analysis**

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score  
less than  $Q_1 - 1.5 \times \text{IQR}$   
or  
more than  $Q_3 + 1.5 \times \text{IQR}$

**Normal distribution**

- approximately 68% of scores have z-scores between  $-1$  and  $1$
- approximately 95% of scores have z-scores between  $-2$  and  $2$
- approximately 99.7% of scores have z-scores between  $-3$  and  $3$

$$E(X) = \mu$$

$$\text{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

**Probability**

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

**Continuous random variables**

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

**Binomial distribution**

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

$$X \sim \text{Bin}(n, p)$$

$$\Rightarrow P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\text{Var}(X) = np(1-p)$$

**Differential Calculus****Function****Derivative**

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

**Integral Calculus**

$$\int f'(x) [f(x)]^n dx =$$

$$\frac{1}{n+1} [f(x)]^{n+1} + c \text{ where } n \neq -1$$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\} \text{ where } a = x_0 \text{ and } b = x_n$$

**Combinatorics**

$${}^n P_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^n C_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1} x^{n-1} a + \dots + \binom{n}{r} x^{n-r} a^r + \dots + a^n$$


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**Vectors**

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1 x_2 + y_1 y_2, \text{ where}$$

$$\underline{u} = x_1 \underline{i} + y_1 \underline{j} \text{ and } \underline{v} = x_2 \underline{i} + y_2 \underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$


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**Complex Numbers**

$$z = a + ib = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

$$[r(\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta) = r^n e^{in\theta}$$


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**Mechanics**

$$\frac{d^2 x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

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Student Number

**TRIAL** HIGHER SCHOOL CERTIFICATE EXAMINATION (BOSS version)

# Mathematics Extension 2

## Section II Writing Booklet

### Instructions

- Use this writing booklet for Section II ONLY
- Write your Centre Number and Student Number at the top of this page
- Follow the instructions inside this booklet as to where you should write your answers
- Write the number of this booklet and the total number of booklets that you have used for this section (eg: **1** of **3**)

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Question Number

**Answer Question 11 on pages 3–6.***Do NOT write in this area.*



**Do NOT write in this area.**



If you require more space to answer Question 11, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 11, tick here.

*Do NOT write in this area.*

**End of Question 11**

**Go to page 8 for Question 12**

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Question Number

**Answer Question 12 on pages 8–11.**

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**Do NOT write in this area.**





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**Do NOT write in this area.**

If you require more space to answer Question 12, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 12, tick here.

**End of Question 12**

**Go to page 13 for Question 13**

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Question Number

**Answer Question 13 on pages 13–16.***Do NOT write in this area.*



*Do NOT write in this area.*



***Do NOT write in this area.***



If you require more space to answer Question 13, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 13, tick here.

*Do NOT write in this area.*

**End of Question 13**

**Go to page 18 for Question 14**



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Question Number

**Answer Question 14 on pages 18–21.**

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**Do NOT write in this area.**



**Do NOT write in this area.**

If you require more space to answer Question 14, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 14, tick here.

**End of Question 14**

**Go to page 23 for Question 15**

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Question Number

**Answer Question 15 on pages 23–26.***Do NOT write in this area.*



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If you require more space to answer Question 15, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 15, tick here.

*Do NOT write in this area.*

**End of Question 15**

**Go to page 28 for Question 16**

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Question Number

**Answer Question 16 on pages 28–31.**

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**Do NOT write in this area.**



**Do NOT write in this area.**

If you require more space to answer Question 16, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 16, tick here.

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